

GlassBox -- Business Proposal

Accenture Innovation Challenge 2026, Track 3

Team Vantage -- Accenture Innovation Challenge 2026, Track 3

Business Proposal

In one sentence: GlassBox turns a three-to-four-day manual investigation into a cited, confidence-graded answer in under 30 seconds — and, measured rather than claimed, it is the version of this category of tool that says *"I don't know"* instead of inventing a cause when the data doesn't support one.

Every technical claim in this document is pulled from a committed file, not asserted — the same discipline `eval/scorecard.md` and `CHANGELOG.md` hold the engine to. Where a section instead depends on something no prototype can prove — market size, willingness to pay, positioning against a named incumbent — we say so plainly rather than dressing an estimate up as a finding. That distinction is marked explicitly throughout, starting in §2.

1. The user

Primary: the category or functional manager staring at a number she can't explain. In the prototype that's "Priya," Audio category lead at a mid-size electronics retailer (`README.md`) — no SQL, no analytics degree, and accountable for a KPI that moved before she knew why.

Secondary: the analyst who currently does the investigation by hand. GlassBox does not remove this role — it removes the three-to-four-day mechanical part (pull the numbers, slice by every dimension, read forty tickets) so the analyst's time goes to the last 20%: judgement calls the system correctly declines to make — a real movement with no true cause (scenario `SC-08`) and a confounder that survives every control-segment test (scenario `SC-12`); see `docs/03_SCENARIOS.md` for the full 17-scenario set this repo's evaluation is built on.

Tertiary: Finance/Ops leadership, who receive the action object and the audit trail, not the investigation.

Explicitly not the buyer, at least not first: the data platform team. Adoption cost sits with them (writing semantic contracts), but the person who feels the pain and requests the tool is the KPI owner, not the platform team — this shapes the go-to-market motion in §4.

2. The problem, sized honestly

Measured, not estimated: - The manual investigation this replaces (pull numbers → slice dimensions → ask other teams → read support tickets → produce a written answer) is a real, described workflow, not a hypothetical (`README.md`'s Priya scenario). - GlassBox runs the same seven-stage investigation end to end in **under 30 seconds** measured (`eval/cost_receipt.md`: p50 latency ~424-880ms *per pipeline stage set*, full harness across 17 scenarios in ~26s total, meaning each individual investigation completes in low single-digit seconds to tens of seconds depending on retrieval cold-start — see the receipt for the real breakdown), at **\$0 marginal cost in replay mode** and a small, bounded live-mode cost (2 LLM calls per run, hard-capped).

What we refuse to invent — and the single highest-value thing left to do: how many investigations a month a "mid-size retailer" actually runs, what an analyst-day actually costs a specific customer, or what this is worth to them in rupees. Those numbers need a discovery call with 3-5 real prospective customers, not a benchmark run — so rather than fill the gap with a plausible-sounding placeholder, we're naming it as the first thing a design partner buys us (§4, Phase 1) — see `README.md`'s equivalent note on the same question.

Why this is structural, not seasonal: any organization running weekly/monthly KPI reviews across more than a handful of segments has this bottleneck, because the translation from "what changed" to "why" is inherently manual today — not a tooling gap any current dashboard closes. `docs/05_DATA_STRATEGY.md` §2 makes the case in full: no public dataset even pairs the four things — KPIs, customer text, anomaly labels, cause labels — needed to build or evaluate a system like this, which is itself evidence nobody has solved it cleanly yet.

3. What GlassBox actually is, in procurement terms

Not a chatbot over a warehouse. Not an agent. A **governed, deterministic investigation pipeline** with two narrow, capped LLM touchpoints, sitting behind a semantic contract layer that doubles as the access-control layer.

The reason this framing matters commercially: it is the difference between "another AI pilot that needs a security review nobody has time for" and "a tool where the access control is provably the same as the warehouse's own row-level security, because it's compiled from the same contract." See [docs/06_SECURITY_TRUST.md](#) and the entitlements demonstration in [JUDGES.md](#)'s 15-minute path (a real, run-not-simulated [entitlements_hash](#) diff between personas).

Deployment model: ships to the customer's data, never the reverse — runs inside the customer's own cloud tenant, reads their warehouse through a read-only service account, calls the model through a private endpoint in their own subscription ([README.md](#) §Trust and deployment). This is a deliberate, load-bearing constraint on the product, not an aspiration: it is what makes "can I trust this with my data" answerable in one sentence instead of a security questionnaire.

4. Phased roadmap

Phase 0 (this submission). Engine, evaluation harness, and evidence real and reproducible for 5 semantic contracts and 17 pre-registered scenarios. UI thin but functional ([app/main.py](#)). B1 baseline run; B3 baseline run live against a real model ([eval/baseline_scorecard.md](#)) — not simulated.

Phase 1 — design partner (1 customer, 5 metrics, 90 days). The five metrics that already appear in that customer's own monthly business review, not a generic starter set — this is deliberate: writing a contract is an afternoon with the metric's *owner* ([JUDGES.md](#)), and picking metrics nobody actually reviews wastes that afternoon on a metric nobody will ever ask "why did this move." Success criteria agreed with the partner up front, not retrofitted: hallucinated-cause rate on their real data (target: what we measured here, 0), and a comparison against however they currently do this investigation (their own B1-equivalent), not against our own baseline.

Phase 2 — vertical expansion (3-5 customers, same vertical). Electronics/ general retail first, because Meridian is modeled on it and the contract templates transfer directly. Contracts stay bespoke per customer (the adoption cost is real, see §5), but the *stages* — detect, localize, decompose, retrieve, falsify — do not change per customer; only the contracts and the document corpus do.

Phase 3 — platform. Self-service contract authoring (a governed metric still needs an owner and a review, but the YAML-writing itself doesn't need us in the room), a real vector/hybrid index once corpus size crosses the point where in-memory retrieval stops winning on recall ([docs/adr/0002](#) names the measured threshold reasoning, $\sim 10^5$ - 10^6 documents), and the persona set expanding past CFO/Category Manager/Analyst as real customers tell us which roles actually use it.

What is explicitly NOT on this roadmap, and why: an agent framework (ADR-0003 — the control-flow-must-be-identical-every-run argument is a product requirement, not a current limitation to grow out of), a general-purpose "ask anything about the business" chatbot (the contract-scoped vocabulary and the clarify-rather-than-guess behavior are the trust mechanism; a general chatbot on the same warehouse would need to rebuild it from scratch), and fine-tuning (customer data never needs to leave the tenant to make this work — training on it would contradict the whole deployment story, per [docs/02_BUILD_BLUEPRINT.md](#)'s deliberate-non-features table).

Who buys it, and how: the CDO or analytics platform owner, used day to day by category managers, regional heads and finance controllers — a layer on the BI stack the enterprise already owns, not a rip-and-replace, sold one design partner at a time (Phase 1) before any vertical or platform motion. We are not asserting a price here — that number needs the same discovery conversation named in §2 — but the buyer and the motion are not a gap the way the price is.

5. Risks and mitigations

Risk	Real, or hypothetical?	Mitigation
Contract authoring cost blocks adoption	Real — five metrics is an afternoon each; 400 is a programme (JUDGES.md)	Phase 1 deliberately scopes to 5 metrics with a design partner, not a big-bang rollout. Most enterprises already have half a contract in an existing dbt/semantic layer to draw from.
Synthetic-primary-dataset objection undermines evaluation credibility	Real, and named rather than hidden	We had to build this: no public dataset pairs KPIs, customer text and cause labels (§2 above; full argument in docs/05_DATA_STRATEGY.md §2). So the ground truth in data/injection_manifest.yaml was frozen before the engine existed — provable with git log --follow data/injection_manifest.yaml — and the generator (data/generate.py) is built to make it hard rather than convenient: ramped changes instead of clean steps, partial spillover into the "control" segments, a deliberately confounded rival cause (SC-01's national promo). RS external benchmark (135 real anomalies, 7 published algorithms) adds real-data coverage on top of this — planned, tracked openly in CHECKLIST.md, not yet run.
A confounder survives every control-segment test and produces an overconfident answer	Real — this is test scenario SC-12's designed failure mode	The system's actual behavior on this exact case, measured: it abstains rather than force an answer (eval/scorecard.md's SC-12 row). The tier ladder and falsification stage exist specifically so this degrades honestly instead of silently.
LLM provider dependency / vendor lock-in	Partially mitigated by design, not yet proven	engine/llm_client.py is the single boundary; provider is config, not code. This session added a second working adapter (Groq, alongside the original Anthropic one) without touching any other file — real evidence the vendor-neutral design works, not just an architecture diagram claiming it does.

Risk	Real, or hypothetical?	Mitigation
A live-mode cost surprise at scale	Low, and measured	Two LLM calls per run, hard-capped in the schema (telemetry.llm_calls.maximum: 2) and enforced in code (LLMCallCapExceeded). eval/cost_receipt.md reports real per-run token/cost telemetry every time the harness runs, not on request.
Market size / willingness to pay unvalidated	Real, explicitly not hidden	Named in §2 above as the top open item. No number is asserted for it here.
"Just use one LLM prompt" competitive pressure	Real — this is the most common objection this category of product faces	eval/baseline_scorecard.md's B3 baseline is not hypothetical; it was run live against a real model this session. On test scenario SC-08 specifically it produced a fluent, 70-80%-confident, fabricated cause — quoted verbatim in that file, not paraphrased.

6. The pitch in one paragraph

A KPI moves. Today, finding out why costs three to four analyst-days and finishes after the decision window closes, or it costs one AI prompt and produces a confident, unverifiable, sometimes fabricated paragraph. GlassBox is neither: five of seven stages are arithmetic and statistics with no model in the loop, the two touchpoints that do use one are capped and number-validated, and — measured, not claimed — it produces zero hallucinated causes across 15 scored, pre-registered scenarios (17 pre-registered; two retired in place when their own ground truth turned out to need fixing, published in full rather than quietly dropped) while a real, live run of the obvious one-prompt alternative invents one on the scenario designed to test exactly that. The hard engineering problem was not finding causes. It was building something that stays quiet when there isn't one.